

Enterprise Resource Planning (ERP) System.

ABSTRACT

The Enterprise Resource Planning (ERP) System is an integrated web-based application designed to manage and automate major business activities through a single platform. The system includes modules such as Human Resources, Inventory, Customers, Sales, Accounting, Purchase Orders, Users, Audit Logs, and Reports. It helps organizations maintain centralized data, reduce manual work, improve accuracy, and access business information efficiently. The system also provides features such as account creation, login authentication, password reset, dark/light mode, data management, PDF report generation, and audit tracking. Overall, the ERP system provides a simple, user-friendly, and efficient solution for managing business operations.

INTRODUCTION

Enterprise Resource Planning (ERP) is a software approach used to integrate and manage different business processes within an organization. In traditional systems, business information is often maintained separately, which can lead to duplicate data, manual work, and difficulty in accessing updated information.

The proposed ERP System provides a centralized platform where multiple business activities can be managed from one application. It combines HR Management, Customer Management, Inventory, Sales, Accounting, Purchase Orders, Reporting, User Management, and Audit Logs. The system is designed with a simple and responsive interface so users can easily add, view, update, and remove records. It also provides reporting and PDF export facilities to support business analysis and decision-making.

EXPLANATION

The Enterprise Resource Planning (ERP) System is developed as a single-file web application using HTML, CSS, and JavaScript. HTML is used to create the structure of the website, CSS provides the visual design, responsive layout, colors, animations, and dark/light mode, while JavaScript controls the main functionality, data processing, authentication, and interaction between different modules. The system begins with a login and account-creation interface where users can register or enter their existing credentials. When the entered username and password are validated successfully, the system creates a user session and opens the main ERP dashboard. If the credentials are incorrect, an error message is displayed. The dashboard provides an overview of the organization and gives users access to all major ERP modules.

The HR Management module is used to maintain employee information such as name, department, position, email, and salary. Users can add and remove employee records, and the information is stored in the ERP database. The Customer Management module manages customer records and allows users to add, view, and remove customer information. The Inventory module maintains product details and stock quantities. When a product is involved in a sale, the system can identify the matching inventory product and update its available stock.

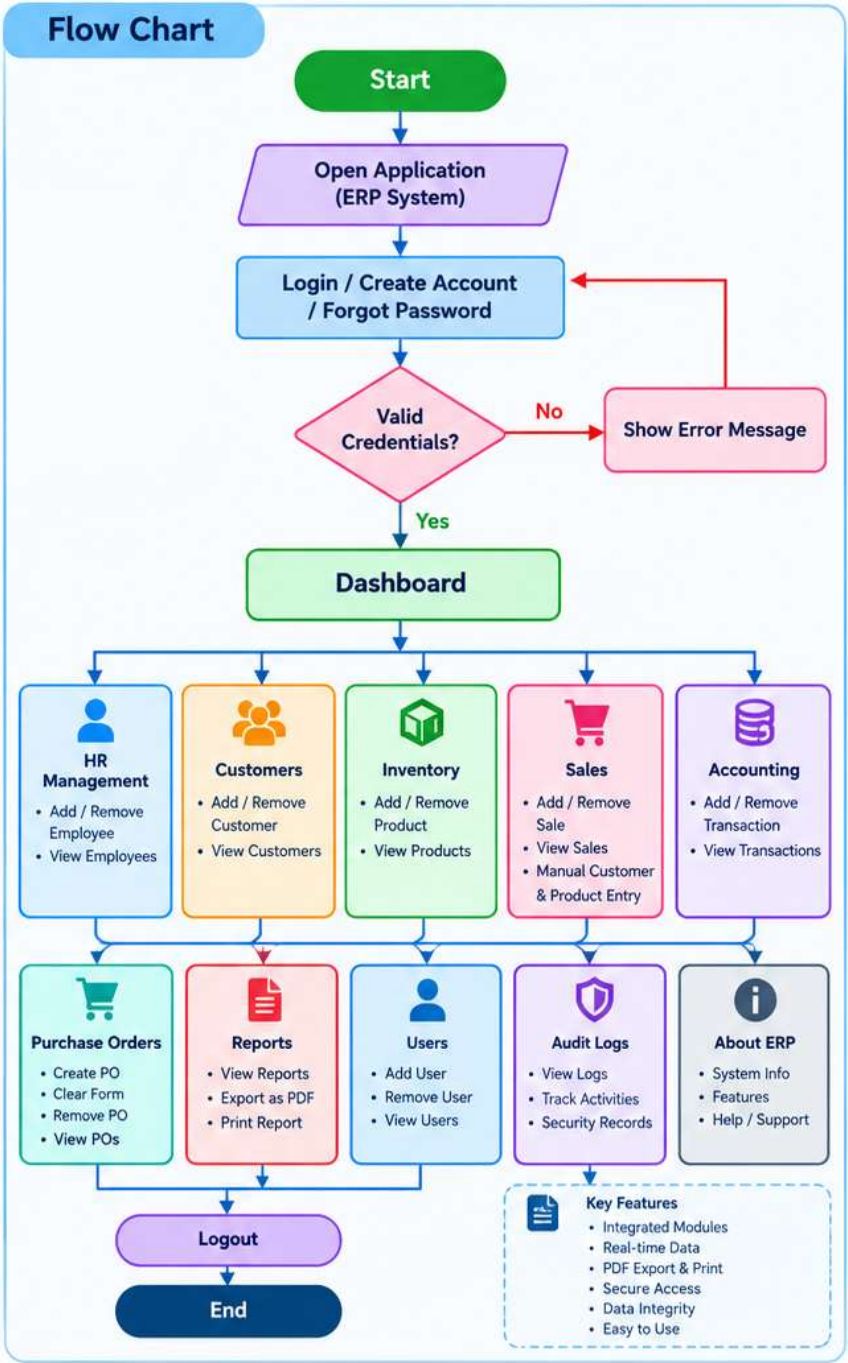
The Sales module allows users to manually enter customer and product names, quantity, and unit price. JavaScript calculates the total sales amount automatically and creates a sales record. If the entered product matches an existing inventory item, the available stock is reduced accordingly. The sale is also recorded as an income transaction so that it can be included in the accounting and reporting sections. The Accounting module manages financial transactions such as income and expenses. Sales can generate income transactions, while purchase orders can generate expense transactions, allowing the system to calculate total income, expenses, and balance.

The Purchase Order module allows users to enter supplier details, product information, quantity, and purchase amount. The system generates a purchase-order number and stores the purchase information. When an amount is entered, the system can also create an expense transaction associated with the purchase. The Reports module collects information from the different ERP modules and generates business summaries, including sales, inventory, income, expenses, balance, employees, customers, purchases, and accounting information. The reports can be viewed, printed, or exported as PDF files.

The system uses browser localStorage to store demonstration ERP data and sessionStorage to maintain information about the currently logged-in user. This allows information entered in one module to be accessed by other modules within the application. The system also includes user management and audit logs to manage users and record important activities such as login, logout, and business operations. Additional features include account registration, forgot-password and reset-password functionality, form validation, search, notifications, responsive design, animations, and dark/light mode.

Overall, the code follows a simple and integrated flow in which the user logs in or creates an account, accesses the dashboard, selects an ERP module, performs operations such as adding, viewing, or removing records, stores the changes, and uses the updated information for reports and business analysis. The combination of HTML, CSS, and JavaScript in a single file provides a simple demonstration of how an ERP system can integrate multiple business functions into one centralized platform.

Flow chart



CONSULSION

The Enterprise Resource Planning (ERP) System provides an integrated solution for managing important business operations in one application. By combining HR, customers, inventory, sales, accounting, purchase orders, reporting, and audit management, the system reduces manual effort and improves accessibility of business information.

The project demonstrates how different business functions can be connected through a centralized ERP platform. Features such as authentication, data management, sales processing, inventory updates, PDF reporting, and audit tracking make the system useful for demonstrating modern business-management concepts. Overall, the ERP System provides a simple, organized, and efficient platform for managing business processes.